

Visualising Household Deprivation in England and Wales: A Comparative Analysis of the 2011 and 2021 Censuses

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1 Abstract

This report examines household deprivation across 175 local areas in England and Wales, using data from the 2011 and 2021 censuses. Six types of data are combined: poverty measures, general health, work activity, education levels, central heating, and number of bedrooms, to study the unseen patterns of social and economic hardship. A PCA analysis of five key variables shows that 74.6% of the differences can be explained by two main factors: general poverty (poor health, unemployment, low education) and a separate rural fuel poverty factor (lack of central heating). Another method, t-SNE, shows clear groups like post-industrial towns, Welsh valleys, London boroughs, and wealthy southern counties. A Bayesian Beta-Binomial method applying Empirical Bayes provides reliable ranges for poverty rates, showing that all 175 areas improved from 2011 to 2021, with inner London areas improving the most (Newham dropped from 75.0 to 60.7, a 14.3-point drop). A Bayesian Linear Regression model predicts 2021 poverty from 2011 data with high accuracy ($R^2 = 0.93$), finding that education, poor health, no central heating, and 4+ bedrooms are significant predictors, while unemployment is not. Nine areas fall outside the expected range, with Brighton and Hove doing worse (+3.3 points) and City of London doing better (-5.6 points) than expected. Analysis shows that the link between unemployment and health changed substantially between the two censuses (correlation ranging from 0.55 to -0.04), likely due to COVID-related changes in the job market. A user test with five people showed perfect accuracy for lookup and pattern tasks, but understanding uncertainty required more explanation.

2 Introduction

Household deprivation quantifies economic and social hardship by measuring what families lack in jobs, education, health, and housing. The UK census scores households from 0 (no lack) to 4 (lacking in all areas). Identifying where deprivation occurs, understanding its causes, and tracking changes over time is crucial for informed policy decisions.

The Index of Multiple Deprivation (IMD) produced by MHCLG (2019)[1] is the main way to measure deprivation across England, combining seven factors. However, unlike the census, the IMD relies on government records and cannot be directly compared across census years. Townsend (1987) [2] first suggested measuring deprivation using census data, focusing on factors such as unemployment, overcrowding, and car ownership, laying the groundwork for the approach this report builds on. Dorling et al. (2007) [3] showed that deprivation in areas of Britain remains largely unchanged over many years, and this report checks that by comparing 2011 and 2021 census data. Building on these methods, this report addresses three questions relevant to a policy analyst or local government decision-maker working on social inequality: How does household deprivation map across England and Wales, and how has this changed from 2011 to 2021? Policymakers must grasp not only where deprivation is severe but also where it has improved or persisted to allocate

funding strategically.

What hidden patterns lie within deprivation? Do primary causes health, jobs, education, housing shift independently, or do they cluster? If deprivation connects, addressing one area could drive broader improvements. If it separates, targeted policies per area become essential.

Can 2011 social and economic conditions predict 2021 poverty levels, and which local authorities differ from these predictions? Areas that are poorer than expected based on their 2011 data have been “left behind” by national progress; areas that are better off than expected have gained more from investment or economic changes.

The intended users of this visualization are government policy analysts and regional planning officers who need to find areas with ongoing poverty, assess how well past efforts worked, and decide where to focus future resources. These users understand statistics but need visual tools to explain results to non-technical audiences, such as elected officials and community leaders. The dashboards help both with exploring data to find patterns and checking specific ideas about individual local authorities.

The analysis uses data from the ONS Census 2011 and 2021 at the county and unitary authority levels (175 matching areas after boundary adjustments). This level of detail gives a national view while still showing regional patterns. Smaller areas, such as Lower Super Output Areas (LSOAs), would allow neighborhood-level studies but are outside the scope of this report.

3 Data Preparation and Abstraction

3.1 Source Data

Six datasets were extracted from the ONS Census for both 2011 and 2021, a total of twelve source files:

- Households by deprivation dimensions: count of households deprived in 0, 1, 2, 3, or 4 dimensions
- General health: self-reported health status (very good to very bad)
- Economic activity: employment status, including unemployed and long-term sick
- Highest level of qualification: from no qualifications to Level 4+
- Central heating: type of heating or absence thereof
- Number of bedrooms: household size proxy

All datasets are organized by county or unitary authority. The 2011 files start at header row 8, and the 2021 files start at header row 7, showing different ONS formatting styles. The data mainly includes numbers (counts and calculated percentages) with the local authority name as the category label.

3.2 Cleaning Steps

1. **Removed regional aggregates.** Both census files include summary rows for regions and national totals. These 14 aggregate categories were filtered out, leaving only individual local authorities.
2. **Removed qualification prefix.** The 2011 qualification file adds a code before local authority names (e.g., “ua-county09:Bedford”). These codes were taken out using a pattern-matching method.
3. **Combined health categories.** “Bad health” and “Very bad health” were combined into a single “Poor health” category for easier year-to-year comparison. The distinction between bad and very bad was less useful than tracking overall poor health rates.
4. **Made economic activity consistent.** The 2011 and 2021 files use different detailed categories. Employment types (full-time, part-time, and self-employed) were combined into a single “Employed” total. The 2021 file separates working students from other active groups, so they were combined.
5. **Made central heating consistent.** The 2021 file contains specific fuel types (e.g., mains gas only, electric only) while the 2011 file uses broader groups. Gas, electric, oil, solid fuel, and “two or more types” were matched to common names for both years.
6. **Made bedroom categories consistent.** The 2011 file has separate columns for “4 bedrooms” and “5 or more bedrooms”, which were combined into “4 or more bedrooms” to match the 2021 file. The 2021 file does not have a “No bedrooms” category, so it was set to zero.

3.3 Boundary Changes

Census area boundaries shifted between 2011 and 2021. The ONS Census Maps resource[4] specifically documents these adjustments.

Three boundary changes were addressed.

1. Bournemouth and Poole were merged into *Bournemouth, Christchurch and Poole* (counts summed).
2. Cumbria was apportioned into Cumberland (66.7%) and Westmorland and Furness (33.3%) using population-based ratios.
3. Northamptonshire was apportioned into North Northamptonshire (40.2%) and West Northamptonshire (59.8%).

Apportionment preserves percentages: successor authorities inherit their parent’s 2011 rates, which approximates uniform deprivation across the parent area. Rhondda Cynon Taff was renamed to Rhondda Cynon Taf.

This results in 175 local authorities common to both years, and 4 additional authorities present only in 2021.

3.4 Derived Variables

All raw numbers were changed to percentages of the relevant total (total households for deprivation, heating, and bedrooms; total usual residents for health, economic activity, and qualifications).

A deprivation summary measure was computed:

- **Deprivation rate** = $100 - \text{Not Deprived } \%$, giving the overall percentage of households deprived in at least one dimension.

3.5 Missing Value Audit

The merged dataset contains 350 rows (175 LAs $\times$ 2 years) and 16 columns. Of 5,600 cells (350 $\times$ 16), none are missing (0.00%). No imputation was needed. Census data at the county/unitary level is complete, unlike LSOAs, where disclosure control causes missingness.

3.6 Feature Selection for Dimensionality Reduction

Seven key variables were evaluated for overlap using Python. Three pairs showed correlation above 0.7:

- Poor Health & Long-term Sick: $r = 0.96$. Long-term sickness is a subset of poor health.
- No Qualifications & Level 4+: $r = -0.86$. These mirror the same education dimension.
- Poor Health & No Qualifications: $r = 0.76$. Both are retained, as each captures distinct deprivation domains.

After selecting key features, the dimensionality reduction process resulted in five variables, each covering a separate domain. These features were normalized to ensure fair comparisons before applying PCA.

3.7 Data Abstraction Summary

Using Munzner's framework [5], the dataset is a simple table: local authorities described by percentages and year. PCA and t-SNE outputs represent positions in a reduced space. Bayesian analysis adds credible intervals. Location is provided as latitude and longitude linked to ONS codes.

4 Task Definition

To classify these tasks, Munzner's action-target framework is used, focusing on distinct analysis goals.

Geographic distribution. What is the spatial pattern of deprivation across England and Wales? Dashboard 1 (Deprivation Overview) encodes deprivation % as color at geographic positions, enabling the user to identify regional patterns (north-south divide, Welsh valleys, coastal pockets). The year filter supports temporal comparison. This is a *synoptic* task; the user surveys the entire dataset to identify high-level patterns.

Comparing dimensions. What is the relative severity of each deprivation dimension? The bar chart in Dashboard 1 decomposes overall deprivation into its constituent dimensions (1-dim, 2-dim, 3-dim, 4-dim) and shows each driver's national average. When linked to the map via filter action, it answers: "For a selected LA, which dimensions drive its deprivation?"

Identifying changes over time. Which LAs improved most and least between 2011 and 2021? Change Over Time uses a diverging colour palette on the posterior mean difference, while the slope charts show individual LA trajectories. The dimension change chart shows which specific domains drove national improvement.

Discovering latent structure. Do LAs form natural groups based on their socio-economic profile? Dashboard 2 (Dimensionality Reduction) presents PCA and t-SNE scatter plots to determine whether deprivation follows a coherent latent structure or is randomly distributed. The loadings chart decodes the PCA axes so the user can interpret what each position means.

Identifying outliers. Which LAs deviate from structural predictions? Dashboard 3 (Predictive Model) plots actual vs. predicted deprivation, with credible interval bands indicating statistically significant deviations. The residual bar chart ranks these outliers. This is a *lookup* task with a derived attribute (residual) as the query target.

5 Visualization Justification

5.1 Choropleth Map (Dashboard 1)

The choropleth map uses a green-to-red colour scale to show deprivation scores by location, based on latitude and longitude. Cleveland and McGill (1984) [6] suggest that map position shows information best, and colour brightness works well for categories and ordered groups.

The colour scale highlights deprivation's middle point, the national average. Areas below average are green; areas above are red. This uses the brain's quick recognition of colour differences [7]. Red marks high deprivation as a warning, matching users' natural associations [5, Ch. 10].

A filled polygon map with ONS boundary shapes (Office for National Statistics geographic divisions) was an option. It was not used because the polygon size shows land area, not population. Large rural areas stand out more, but they have fewer people than smaller city areas [5, Ch. 8].

All visuals use percentages instead of raw counts. Comparing raw counts across areas of different sizes (942 to 648,000 households) would be confusing.

5.2 Deprivation Dimensions Bar Chart (Dashboard 1)

The horizontal bar chart shows deprivation percentages by bar length. Cleveland and McGill (1984) [6] rank bar length just below position in terms of data clarity. Bars are grouped by category and sorted by value, making information easy to find.

The bar chart is connected to the map using Tableau's filter

feature. Shneiderman’s (1996)[8] visual information approach is followed: overview first, then zoom and filter, and finally, details on demand. The map provides the overview; clicking a local area filters the bar chart to display that area’s breakdown.

5.3 Slope Chart and Change Map (Dashboard 2)

The slope chart links 2011 and 2021 values for each category with lines. This visual approach makes use of the human ability to notice slopes, making it easy to identify areas that improved most (steep downward lines) or remained unchanged (flat lines) [6].

The change map uses a blue-to-red colour scale centred on the middle improvement value. Blue indicates the most improvement, while red indicates the least. These colours remain easy to distinguish for colourblind users when using the Brewer RdBu palette [9]. The map enables users to quickly identify areas of great or low improvement [7, Ch. 5].

5.4 PCA Scatter Plot with Loadings (Dashboard 3)

The PCA scatter plot maps the first two principal components to x and y , with deprivation % shown by colour. This design has three justifications.

First, mapping position on a shared scale shows data best [6], aligning with the main dimensions. Second, a smooth colour gradient lets users check whether PC1 matches deprivation by observing whether green-to-red colours shift left to right.

Third, the loadings bar chart next to the scatter plot helps users understand. PCA axes are abstract; without loadings, users cannot know what “high PC1” means. The chart uses bar length to indicate variable contribution, with blue for positive and red for negative, corresponding to the scatter plot colours. Users see “No Qualifications” as a positive loading on PC1 and understand that points on the right have high no-qualification rates.

Tooltips on the PCA and t-SNE scatter plots show local area name, year, and deprivation percentage. This helps users see the original set of variables, letting them interpret positions in familiar terms rather than abstract values.

PCA summarises data by keeping as much variation as possible [10]. For this data, two components capture 74.6% of the variation in five variables, giving a clear summary. PCA was chosen over other methods because the goal is to show variation structure, not hidden variables. Data were standardised using z -scores before PCA because the baseline scales varied (No Central Heating: 0–14%; No Qualifications: 5–35%). Without standardisation, variables with bigger ranges would dominate the components.

The Kaiser rule (eigenvalues > 1) decided the number of components. PC1 (2.60) and PC2 (1.13) were kept; PC3 to PC5 (under 1) were dropped. Analysis showed PC1 loadings were stable across census years (Poor Health: 0.527–0.529, No Qualifications: 0.498–0.564). This supports using the same PCA space to compare years, even though the link between Unemployment and Health changed substantially ($r = 0.554$ in 2011, $r = -0.038$ in 2021).

5.5 t-SNE Scatter Plot (Dashboard 3)

t-SNE [11] preserves local neighbourhood structure, complementing PCA’s focus on global variance. With PCA, we see a gradient from affluent to deprived. t-SNE shows discrete clusters, highlighting groups of LAs that differ across all five features and stand apart from others.

The same colours (deprivation score) and shapes (circle or hollow for year) are used as in the PCA scatter. This sustains the dashboard’s consistency and is easy to follow, since the design does not change [5, Ch. 5].

UMAP was considered because it is faster, but t-SNE was chosen. The dataset is small (350 points), and t-SNE focuses on local groups, helping find clusters. t-SNE was run on the top 3 PCA components, following advice to reduce data size first for performance [?]. Perplexity was set to 40, about the square root of the sample size ($\sqrt{350} \approx 18.5$, rounded up for two years), within the suggested range of 5–50. Stability was assessed by running t-SNE with perplexities of 20, 30, 40, and 50; the cluster patterns remained consistent across perplexities.

One drawback of t-SNE is that spaces between groups do not really mean anything; only closeness within a group is clear. This is explained in the dashboard notes to prevent confusion.

5.6 Bayesian Prediction Scatter (Dashboard 4)

The actual-vs-predicted scatter plot shows predicted values on the x -axis and actual 2021 values on the y -axis. The $y = x$ line shows perfect prediction. Points above the line are “left behind” (more deprived than predicted), and points below “overperformed.” This design uses people’s tendency to notice differences from a straight line. Even small differences stand out [6].

Red and green colours highlight outliers, using the pop-out property. Nine statistically meaningful deviations (points outside the 95% CI bands) stand out against the grey mass of well-predicted LAs. The residual bar chart on the right ranks these outliers, providing another view. The scatter shows spatial patterns in prediction error; the bars support precise value comparison.

The Bayesian Linear Regression model uses conjugate Normal-Inverse-Gamma priors, which allow closed-form (analytical) solutions for the posterior distribution and avoid the need for MCMC (Markov Chain Monte Carlo) sampling, a slower method for assessing uncertainty. Model results are summarised in Table 1.

Table 1: Model comparison — Bayesian Linear Regression outperforms Random Forest while providing calibrated credible intervals (94.9% coverage).

Model	R ²	MAE	RMSE	Has CI
Baseline (Mean)	−0.013	3.934	4.913	No
Bayesian Linear Regression	0.928	1.011	1.308	Yes
Random Forest	0.881	1.221	1.681	No

The posterior coefficients (Table 2) reveal that only 2011 deprivation rate ($\beta = 4.238$), no qualifications, poor health, no central heating, and 4+ bedrooms are significant. Unemploy-

ment is not significant after controlling for past deprivation, consistent with the structural decoupling identified in the correlation stability analysis.

Table 2: Bayesian posterior coefficients for the levels model. Only unemployment is non-significant; all other features predict 2021 deprivation.

Feature	Posterior Mean	95% CI	Significant
Intercept	52.349	[52.168, 52.529]	Yes
Poor Health %	-0.441	[-0.820, -0.062]	Yes
Unemployed %	-0.064	[-0.424, 0.295]	No
No Qualifications %	1.595	[1.259, 1.930]	Yes
No Central Heating %	-0.231	[-0.455, -0.008]	Yes
4+ Bedrooms %	0.332	[0.011, 0.653]	Yes
2011 Deprivation Rate	4.238	[3.754, 4.721]	Yes

6 Evaluation

Using Munzner’s four-level framework, each part of the model was evaluated using appropriate validation methods.

6.1 Domain Situation Validation

Threat: The dashboard may address an incorrect problem. All five group members agreed that the dashboard addresses the main questions about household deprivation. Two suggested extensions, including more detailed geographic breakdowns and funding allocation analysis. This confirms that the domain situation is well captured, while also highlighting some scope limitations.

6.2 Data/Task Abstraction Validation

Threat: The dashboard may display inappropriate data or tasks.

Participants were asked whether the five driver variables adequately represent the four deprivation dimensions. Three noted that income is missing, meaning housing-related insights rely on indirect measures. One participant questioned whether the *4+ bedrooms* variable indicates deprivation or wealth. Overall, feedback supports the feature set but highlights that some variables act as proxies rather than direct measures.

6.3 Visual Encoding/Interaction Validation

Threat: The visual encoding may lack effectiveness.

Five task-based evaluations were conducted with group members:

Lookup: “What is the deprivation rate for Bristol, City of, in 2021?” Mean time: 12 seconds. Accuracy: 5/5. The map tooltip enabled efficient retrieval.

Comparison: “Which improved more between 2011 and 2021 - Manchester or Liverpool?” Mean time: 18 seconds. Accuracy: 4/5 (one misread year). The change map and slope chart supported comparison effectively.

Pattern: “Looking at the PCA plot, do London boroughs cluster together?” Mean time: 8 seconds. Accuracy: 5/5. Participants quickly identified clustering patterns, indicating effective visual encoding.

Principal Component Analysis (PCA) reduces dimensionality by combining multiple variables into key components that capture most variance.

Uncertainty Interpretation: Three out of five participants correctly interpreted uncertainty intervals, while two identified the widest interval but could not explain it. This indicates that the relationship between population size and uncertainty is not visually clear.

Outlier Detection: “Can you find any LA that seems unusually positioned in the t-SNE plot?” Mean time: 15 seconds. Accuracy: 5/5 identified the Isles of Scilly; 2/5 also identified the City of London.

t-SNE (t-Distributed Stochastic Neighbor Embedding) is a technique for visualizing high-dimensional data in two dimensions while preserving local structure.

Overall, the visual design performs strongly for lookup and pattern recognition tasks (100% accuracy, under 15 seconds). It is moderately effective for uncertainty interpretation (60% accuracy), suggesting that Bayesian uncertainty requires clearer visual explanation.

6.4 Algorithm Validation

Threat: The dashboard computation may be too slow.

All Tableau dashboards load within 2 seconds, with filter interactions responding in under 0.5 seconds. The Python pipeline (including PCA, t-SNE, and Bayesian cross-validation) completes in under 30 seconds on a standard laptop. No performance bottlenecks were observed.

7 Conclusion

7.1 Socio-Economic Findings

Five findings emerge from this analysis with direct policy relevance. The following sections describe each finding and contextualize its implications.

The first finding emphasizes that national improvement reflects redistribution rather than elimination. Here, ‘redistribution’ means that households shifted between levels of deprivation, rather than deprivation being entirely removed. ‘Deprivation’ refers to households lacking one or more basic necessities. Overall deprivation fell from 58.7% to 52.4% between 2011 and 2021, and all 175 local authorities improved. However, the share of households in the mildest category (experiencing deprivation in only one dimension, such as education, housing, or health) rose slightly from 32.7% to 33.5%, indicating that many households moved from severe to mild deprivation rather than out of deprivation entirely. Policy success has been a leftward shift along the deprivation spectrum, and the ‘last mile’ of moving households out of any deprivation remains the hardest. This sets the stage for understanding more localized patterns.

Building on these national shifts, the second finding examines inner London’s outperformance against the persistence hypothesis. The most deprived inner London boroughs in 2011 (Newham 75.0%, Tower Hamlets 67.3%, Hackney 68.5%) became the fastest improvers by 2021, contradict-

ing Dorling et al.'s (2007)[3] thesis of persistent geographic inequality. Extending the discussion on local improvement, the City of London (-14.6pp) and Newham (-14.3pp) lead by a wide margin over post-industrial towns like Blackpool (-6.6pp). However, this improvement at the LA level may partly reflect gentrification wealthier residents replacing existing deprived ones rather than individual households improving. Census data cannot distinguish these mechanisms.

Turning to structural aspects of deprivation, the third finding shows a coherent two-axis structure. PCA shows that 74.6% of variance is explained by two factors: a general deprivation axis combining health, employment, and education (PC1, 52.1%), and a separate rural fuel poverty axis (PC2, 22.5%, dominated by 'no central heating'). PC1 loadings remained stable across years (Poor Health 0.527 $\rightarrow$ 0.529), validating cross-year comparison. This argues for integrated interventions on PC1 dimensions and a separate, focused policy for fuel poverty.

The fourth finding addresses resistance to educational improvement. No Qualifications fell from 22.9% to 18.5% - a 4.4pp drop, smaller than the 6.4pp drop in overall deprivation. Areas with high baseline No Qualifications were the most resistant to improvement ($\beta = 1.464$ in the change model). This is consistent with education being a generational-scale variable: improving qualification levels requires waiting for cohort replacement.

Examining changes in labor market dynamics, the fifth finding shows that the unemployment-health link broke between censuses, but the basic structure stayed intact. The correlation between unemployment and poor health collapsed from $r = 0.554$ in 2011 to $r = -0.038$ in 2021, plausibly indicating the COVID-19 furlough scheme. Yet PCA structure remained stable, and unemployment is the only feature that fails to independently predict 2021 deprivation in the levels model ($\beta = -0.064$, not significant). The 2021 census, therefore, captured a labor market temporarily separated from its usual systemic relationships.

Methodological triangulation reveals model-dependent outliers. Bayesian Linear Regression and Random Forest agree on City of London as an overperformer but disagree on Newham (Bayesian: green/overperformer; RF: red/left-behind, residual +3.6). Brighton and Hove is distinctive as the only Bayesian-flagged underperformer (+3.3pp), likely reflecting housing-driven compositional change.

7.2 Lessons Learned About Information Visualization

In data visualization, selecting the appropriate representation proved more impactful than visual aesthetics. Comparing choropleth maps and bar charts illustrated Cleveland and McGill's (1984)[6] perceptual hierarchy in practice: colour supports rapid spatial pattern recognition, while length enables precise comparison.

Clear interpretation of derived variables is equally important. PCA scatter plots are difficult to interpret without supporting loadings, as users cannot infer what "high PC1" represents. Pairing scatter plots with loading charts and tooltips transformed the representation from opaque to interpretable, highlighting the importance of multi-layered encoding when

using dimensionality reduction.

Uncertainty communication remains a key challenge. Bayesian credible intervals were correctly interpreted in only 60% of user tests, compared to 100% accuracy for simpler pattern tasks. This indicates that statistical uncertainty requires explicit visual explanation, potentially through annotations or interactive linking to population size and interval width.

Validation is essential rather than optional. Correlation checks between 2011 and 2021 revealed meaningful structural changes, including the unemployment-health decoupling, which would have been missed without cross-year comparison. Reporting such diagnostics improves transparency and credibility.

Finally, interaction is necessary for scalability. Visualising 175 local authorities across two time points without filtering would result in severe clutter. Consistent with Shneiderman's (1996) overview-zoom-filter principle, interactive exploration was essential for managing complexity in the dataset.

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